

RUI ZHOU

+86-133-4988-8373 | zhourui9813@gmail.com | [zhourui9813.github.io](https://github.com/zhourui9813)

 Google Scholar |  Github |  linkedin |  Twitter

EDUCATION

- **The Hong Kong University of Science and Technology** Sep. 2026 – Present
Ph.D. in Artificial Intelligence, advised by Prof. Sirui Han and Prof. Yike Guo Hong Kong, SAR
 - Research Interests: Robotics, Embodied AI, Egocentric Human Data
- **Wuhan University** Sep. 2022 – Jun. 2026
B.Eng. in Electronic Information Engineering Wuhan, China
 - Grade: 3.95/4.00; Ranked 3rd among 203 students (**Top 1.5%**)

RESEARCH PUBLICATIONS

C=CONFERENCE, J=JOURNAL, P=PREPRINTS

- [C.1] Rui Zhou*, Yibo Yuan*, Junkai Zhao*, Fangyuan Zhao, Xiaoguang Zhao, Shanghang Zhang[†], Sirui Han[†] (2026). **DexRoam: Enabling Mobile Dexterous Manipulation from Egocentric Human Demonstrations**. *Conference on Robot Learning (CoRL)* 2026.
- [C.2] Rui Zhou, Jingbin Liu[†], Junbin Xie, Jianyu Zhang, Yingze Hu, Jiele Zhao (2026). **STAG-VIO: Stabilized Prompt-to-Geometry Interface for Robust Dynamic Visual-Inertial Odometry**. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* 2026.
- [C.3] Chengbo Yuan, Rui Zhou*, Mengzhen Liu*, Yingdong Hu, Shengjie Wang, Li Yi, Chuan Wen, Shanghang Zhang, Yang Gao[†] (2025). **MotionTrans: Human VR Data Enable Motion-Level Learning for Robotic Manipulation Policies**. *IEEE International Conference on Robotics and Automation (ICRA)* 2026.
- [C.4] Qinwen Xu*, Jiaming Liu*, Rui Zhou*, Shaojun Shi*, Nuowei Han*, Zhuoyang Liu, Chenyang Gu, Shuo Gu, Yang Yue, Gao Huang, Wenzhao Zheng, Sirui Han, Peng Jia, Shanghang Zhang[†] (2026). **TwinRL: Digital Twin-Driven Reinforcement Learning for Real-World Robotic Manipulation**. *ACM Multimedia (MM)* 2026.
- [P.1] Yueru Jia*, Jiaming Liu*, Shengbang Liu*, Rui Zhou, Wanhe Yu, Yuyang Yan, Xiaowei Chi, Yandong Guo, Boxin Shi, Shanghang Zhang[†] (2025). **Video2Act: A Dual-System Video Diffusion Policy with Robotic Spatio-Motional Modeling**. *arXiv:2512.03044*.

EXPERIENCE

- **Beijing Academy of Artificial Intelligence**  Sep. 2025 - Sep. 2026
Research Intern, Supervised by Prof. Shanghang Zhang Beijing, China
 - Developed **DexRoam**, a complete system for learning mobile bimanual dexterous manipulation from egocentric whole-body human demonstrations, integrating portable capture, explicit whole-body alignment, and real-robot policy learning under a single *continuous, coupled* action interface
 - Built a **tracker-free** egocentric whole-body capture system recording full-body motion, hand articulation, and stereo video with only a consumer VR headset and a head-mounted camera, with in-headset streaming that keeps the demonstrator's view consistent with the policy observation
 - Designed three explicit alignment stages (**embodiment, action-semantic, temporal**) mapping human motion into a unified action space while preserving continuity, making human and robot data interchangeable for VLA policies
 - Validated on 5 real-world tasks and 2 VLA backbones: raised average success from **29%→56%** on GR00T N1.7 and **32%→57%** on $\pi_{0.5}$, matching robot-only training with **50% fewer** robot demonstrations
- **EVAR Lab, IIIS, Tsinghua University**  Dec. 2024 - Aug. 2025
Research Intern, Supervised by Prof. Yang Gao Beijing, China
 - Systematically explored whether human data can enable robot policies to directly learn new motions for task completion, through multi-task human-robot cotraining
 - We propose **MotionTrans**, a framework that enables motion-level learning from VR-collected human data, allowing end-to-end robot policies to directly acquire new motions for task completion.
 - Built the Quest 3 teleoperation system for both the real robot (Franka arm + Inspire hand) and bimanual Isaac Gym simulation, open-sourced as **DexTeleGym** , featuring low-latency WebRTC streaming, real-time IK, dexterous hand retargeting, and gesture-controlled episode recording
 - By cotraining 30 human-robot tasks simultaneously, directly transferred motions of **13** tasks from human data to deployable end-to-end robot policies, with **9** tasks achieving non-trivial success rates in a **zero-shot** manner, and enhanced pretraining-finetuning performance by **+40%** success rate
- **HMI Lab, School of Computer Science, Peking University**  Aug. 2025 - Feb. 2026
Research Intern, Supervised by Prof. Shanghang Zhang Beijing, China

- We propose **TwinRL**, a digital twin–real-world collaborative post-training framework that expands and guides RL exploration for VLA models through SFT warm-up, twin RL warm-up, and real-world RL
- Conducted real-world human-in-the-loop online RL experiments on a 7-DoF Franka FR3; across four tasks, TwinRL achieves near-**100%** success in both in-distribution and OOD regions, with **30%+** faster convergence than prior real-world RL methods using only **20 min** of on-robot interaction
- Contributed to **Video2Act**, a dual-system VLA using a video diffusion model to guide a fast DiT action expert; participated in real-world experiments across 6 tasks, surpassing prior SOTA VLAs by **21.7%** in success rate

PROJECTS

- **DexTeleGym: Immersive VR Bimanual Dexterous Teleoperation in Isaac Gym** [🔗] Dec. 2024 - Jan. 2025
Tools: Python, Isaac Gym, Meta Quest 3, WebRTC, Pinocchio, dex-retargeting
 - Developed an immersive VR teleoperation pipeline in NVIDIA Isaac Gym based on OpenTeleVision, enabling real-time control of dual Franka Panda arms with Inspire dexterous hands using Meta Quest 3
 - Implemented low-latency WebRTC streaming from simulation to the headset, with multi-modal visual feedback
 - Integrated real-time arm IK via Pink/Pinocchio and human-to-robot hand retargeting via dex-retargeting
 - Designed gesture-based controls for episode recording and display switching to support demonstration collection for robot learning; open-sourced with **60+** GitHub stars

HONORS AND AWARDS

- **National Scholarship (Top 0.2% nationwide)** Dec. 2024
- **Wuhan University Excellent Student Class A Scholarship (Top 5% university-wide)** Oct. 2024
- **Wuhan University Huang Zhangren Special Scholarship** Oct. 2023
- **First Prize, Provincial Level, Mathematics Competition of Chinese College Students** Dec. 2023
- **National Training Program of Innovation and Entrepreneurship for Undergraduates** Mar. 2024

SERVICES

- **Reviewer, IEEE International Conference on Robotics and Automation (ICRA)**
- **Reviewer, IEEE Robotics and Automation Letters (RA-L)**
- **Reviewer, IEEE International Conference on Multimedia and Expo (ICME)**